

mb-pc40s-2013-jun

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

A central angle of a circle subtends an arc length of 5π cm.

Given the circle has a radius of 9 cm , find the measure of the central angle in degrees.

Solution

$$s = \theta r$$

$$5\pi = \theta(9) \quad 1/2 \text{ mark for substitution into correct formula}$$

$$\theta = \frac{5\pi}{9} \quad 1/2 \text{ mark for solving for } \theta$$

$$\begin{aligned} \theta(\text{ in degrees }) &= \frac{5\pi}{9} \cdot \frac{180^\circ}{\pi} \\ &= 100^\circ \end{aligned}$$

1 mark for conversion to degrees

2 marks

Question 2

Solve the equation $\csc^2 \theta + 3 \csc \theta - 4 = 0$ over the interval $[0, 2\pi]$.

Express your answers as exact values or correct to 3 decimal places.

Solution

Method 1

$$\begin{aligned}\csc^2 \theta + 3 \csc \theta - 4 &= 0 \\ (\csc \theta - 1)(\csc \theta + 4) &= 0 \\ \csc \theta = 1 &\quad \csc \theta = -4 \\ \sin \theta = 1 &\quad \sin \theta = -\frac{1}{4} \\ \theta = \frac{\pi}{2} &\quad \theta_r = 0.252680\end{aligned}$$

or

$$\theta = 1.570796 \quad \theta = 3.394273, 6.030505$$

2 marks (1 mark for consistent solutions of

each trigonometric equation)

or

4 marks

Question 3

Jess invests \$12000 at a rate of 4.75% compounded monthly.

How long will it take for Jess to triple her investment?

Express your answer in years, correct to 3 decimal places.

Solution

Method 1

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$36000 = 12000\left(1 + \frac{0.0475}{12}\right)^{12t} \quad 1/2 \text{ mark for substitution}$$

$$3 = \left(1 + \frac{0.0475}{12}\right)^{12t}$$

$$\ln 3 = \ln \left(1 + \frac{0.0475}{12}\right)^{12t} \quad 1/2 \text{ mark for applying logarithms}$$

$$\ln 3 = 12t \ln \left(1 + \frac{0.0475}{12}\right) \quad 1 \text{ mark for power rule}$$

$$t = \frac{\ln 3}{12 \ln \left(1 + \frac{0.0475}{12}\right)} \quad 1/2 \text{ mark for isolating } t$$

$$t = 23.174425 \quad 1/2 \text{ mark for evaluating quotient of logarithms}$$

$$t = 23.174 \text{ years} \quad \mathbf{3 \text{ marks}}$$

Question 4

The 4 th term in the binomial expansion of $\left(qx^2 - \frac{3}{x}\right)^{10}$ is $414720x^{11}$.

Determine the value of q algebraically.

Solution

$$t_4 = {}_{10}C_3 (qx^2)^7 \left(-\frac{3}{x}\right)^3 \quad 2 \text{ marks (1 mark for } {}_{10}C_3, 1/2 \text{ mark for each consistent factor)}$$

$$414720x^{11} = 120(q^7x^{14})\left(-\frac{27}{x^3}\right) \quad 1/2 \text{ mark for comparing coefficients}$$

$$414720 = -3240q^7$$

$$q^7 = -128 \quad 1/2 \text{ mark for solving for } q$$

$$q = -2 \quad 3 \text{ marks}$$

Question 5

Bella has 2 pairs of shoes, 3 pairs of pants, and 10 shirts.

Carey has 4 pairs of shoes, 4 pairs of pants, and 4 shirts.

An outfit is made up of one pair of shoes, one pair of pants, and one shirt.

Who can make more outfits? Justify your answer.

Solution

Bella: $2 \times 3 \times 10 = 60$ outfits

Carey: $4 \times 4 \times 4 = 64$ outfits

$\therefore$ Carey can make more outfits.

1 mark for justification

1 mark

Question 6

2 marks

In the binomial expansion of $(x - y)^{10}$, how many terms will be positive?

Justify your answer.

Solution

Six terms will be positive.

The term will be positive when " $-y$ " has an even exponent.

1 mark for six terms

1 mark for justification

2 marks

Question 7

Solve the following equation algebraically where $180^\circ \leq \theta \leq 360^\circ$.

$$2 \sin^2 \theta + 5 \cos \theta + 1 = 0$$

Solution

$$2(1 - \cos^2 \theta) + 5 \cos \theta + 1 = 0 \quad 1 \text{ mark for identity}$$

$$2 - 2 \cos^2 \theta + 5 \cos \theta + 1 = 0$$

$$2 \cos^2 \theta - 5 \cos \theta - 3 = 0$$

$$(2 \cos \theta + 1)(\cos \theta - 3) = 0$$

$$\cos \theta = -\frac{1}{2} \quad \cos \theta = 3$$

$$\theta_r = 60^\circ \therefore \text{no solution}$$

$$\theta = 240^\circ$$

- award a maximum of 3 marks if not solved algebraically

Question 8

Solve the following equation algebraically:

$$\log_3(x-4) + \log_3(x-2) = 1$$

Solution

Method 1

$$\log_3(x-4) + \log_3(x-2) = 1$$

$$\log_3(x-4)(x-2) = 1$$

$$3^1 = (x-4)(x-2)$$

$$3 = x^2 - 6x + 8$$

$$0 = x^2 - 6x + 5$$

$$0 = (x-5)(x-1)$$

$$x = 5$$

1 mark for product rule

1 mark for exponential form

$\frac{1}{2}$ mark for solving for x within a quadratic equation

$\frac{1}{2}$ mark for rejecting extraneous root

3 marks

Method 2

$$\log_3(x-4) + \log_3(x-2) = 1$$

$$\log_3(x-4)(x-2) = 1$$

$$\log_3(x^2 - 6x + 8) = \log_3 3$$

$$x^2 - 6x + 8 = 3$$

$$x^2 - 6x + 5 = 0$$

$$(x-1)(x-5) = 0$$

$$x < 1 \quad x = 5$$

1 mark for product rule

$\frac{1}{2}$ mark for logarithmic form

$\frac{1}{2}$ mark for equating arguments

$\frac{1}{2}$ mark for solving for x within a quadratic equation

$\frac{1}{2}$ mark for rejecting extraneous roots

3 marks

Question 9

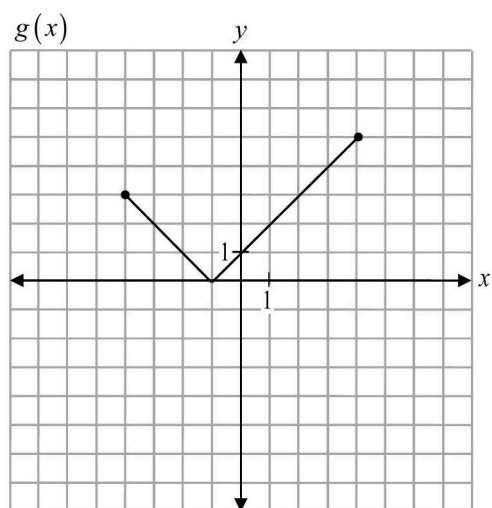
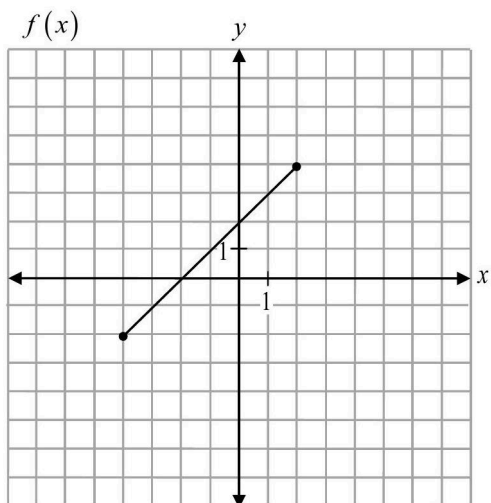
Given that $f(x) = \{(1, 3), (2, 5), (3, 4), (4, 2)\}$, find $f(f(3))$.

Solution

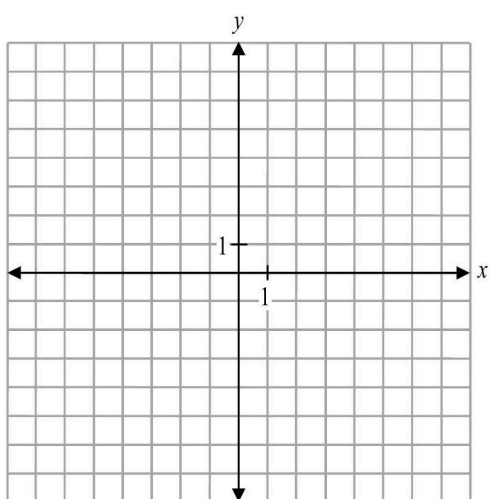
$$\begin{aligned} f(f(3)) &= f(4) \quad 1/2 \text{ mark for } f(3) = 4 \\ &= 2 \quad 1/2 \text{ mark for } f(4) = 2 \\ &\quad \mathbf{1 \textit{ mark}} \end{aligned}$$

Question 10

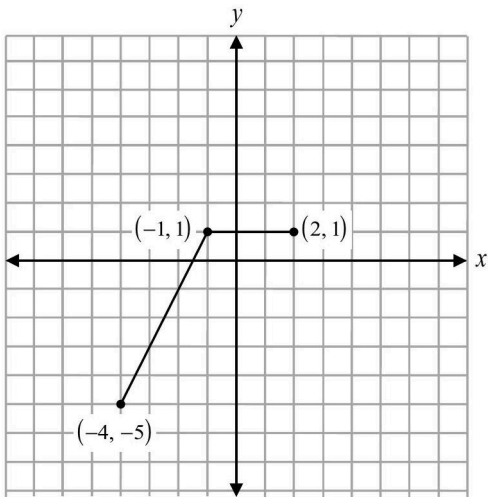
Given the graphs of $f(x)$ and $g(x)$ below,



sketch the graph of $y = f(x) - g(x)$.



Solution



x	$f(x)$	$g(x)$	$f(x) - g(x)$
-4	-2	3	-5
-2	0	1	-1
-1	1	0	1
0	2	1	1
2	4	3	1

1 mark for subtraction of $f(x) - g(x)$

1 mark for restricted domain

2 marks

Question 11

Given the graph of $y = f(x)$, describe the transformations to obtain the graph of the function

$$y = f(2x - 6).$$

Solution

Method 1

Factor out the 2.

$$y = f(2(x - 3))$$

Horizontally compress by a factor of 2.

Then shift 3 units to the right.

1 mark for starting with a horizontal compression

by a factor of 2

1 mark for ending with a horizontal shift of 3 units

to the right

2 marks

Method 2

$$y = f(2x - 6)$$

1 mark for starting with a horizontal shift of 6 units

to the right

1 mark for ending with a horizontal compression by

a factor of 2

2 marks

Question 12

Given $f(x) = \{(-3, 4), (2, 7), (8, 6)\}$, state the domain of the resulting function after $f(x)$ is reflected through the line $y = x$.

Solution

Domain: $\{4, 6, 7\}$ 1 mark for correct domain

1 mark

Note(s):

- award $\frac{1}{2}$ mark for stating the inverse of the function: $f^{-1}(x) = \{(4, -3), (7, 2), (6, 8)\}$

Question 13

Determine the value of y in the following equation:

$$\log_x 27 - \log_x 3 = 2 \log_x y$$

Solution

$$\log_x 27 - \log_x 3 = 2 \log_x y$$

$$\log_x \frac{27}{3} = 2 \log_x y$$

$$\log_x 9 = \log_x y^2$$

$$9 = y^2$$

$$y = \pm 3$$

$$y = 3$$

1 mark for quotient rule

1 mark for power rule

$\frac{1}{2}$ mark for positive value of y

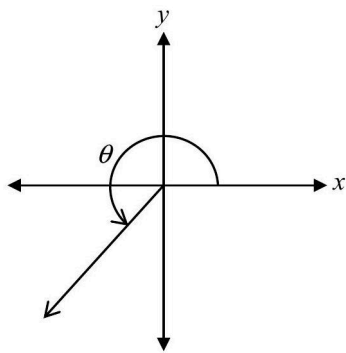
$\frac{1}{2}$ mark for negative value of y and rejecting extraneous root

3 marks

Question 14

Angle θ , measuring $\frac{5\pi}{4}$, is drawn in standard position as shown below.

Determine the measures of all angles in the interval $[-4\pi, 2\pi]$ that are coterminal with θ .



Solution

$$\theta = -\frac{3\pi}{4} \quad 1/2 \text{ mark}$$

$$\theta = -\frac{11\pi}{4} \quad 1/2 \text{ mark}$$

1 mark

Question 15

Prove the identity below for all permissible values of x :

$$\frac{\sin^2 x}{\sec x + 1} = \cos x - \cos^2 x$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

$$\begin{aligned}\text{LHS} &= \frac{1 - \cos^2 x}{\frac{1}{\cos x} + 1} \\&= \frac{1 - \cos^2 x}{\frac{1 + \cos x}{\cos x}} \\&= (1 - \cos^2 x) \left(\frac{\cos x}{1 + \cos x} \right) \\&= (1 - \cos x)(1 + \cos x) \left(\frac{\cos x}{1 + \cos x} \right) \\&= (1 - \cos x)(\cos x) \\&= \cos x - \cos^2 x \\&= \text{RHS}\end{aligned}$$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 16

Solve algebraically:

$${}_nC_2 = 4n + 5$$

Solution

$$n_2 = 4n + 5$$

$$\frac{n!}{(n-2)!2!} = 4n + 5$$

$$\frac{n(n-1)(n-2)!}{\frac{(n-2)!2!}{n(n-1)}} = 4n + 5$$

$$n^2 - n = 8n + 10$$

$$n^2 - 9n - 10 = 0$$

$$(n-10)(n+1) = 0$$

$$n = 10 \quad n \geq -1$$

$\frac{1}{2}$ mark for factorial notation

$\frac{1}{2}$ mark for factorial expansion

$\frac{1}{2}$ mark for simplification of factorial

$\frac{1}{2}$ mark for simplification

$\frac{1}{2}$ mark for both values of n

$\frac{1}{2}$ mark for rejecting extraneous root

3 marks

Question 17

How many different arrangements are possible when arranging all of the letters of the word SEPTEMBER?

A.

$9!$

B.

$6!3!$

C.

$\frac{9!}{3!}$

D.

$\frac{6!}{3!}$

Solution

Correct answer: C

$\frac{9!}{3!}$

Question 18

Which one of the following angles terminates in Quadrant III?

A.

3 radians

B.

$\frac{7\pi}{5}$ radians

C.

-210°

D.

500°

Solution

Correct answer: B

$\frac{7\pi}{5}$ radians

Question 19

There are 13 terms in the expansion of $(3x - y)^{2n}$. Determine the value of n .

A.

6

B.

6.5

C.

7

D.

26

Solution

Correct answer: A

6

Question 20

Which of the following is true about the periods of the three functions below?

$$f(\theta) = 2 \sin 3\left(\theta - \frac{\pi}{2}\right) \quad g(\theta) = \sin 3\theta + 6 \quad k(\theta) = 3 \sin \theta + 6$$

A.

The graphs of $f(\theta)$ and $g(\theta)$ have the same period.

B.

The graphs of $g(\theta)$ and $k(\theta)$ have the same period.

C.

All of the graphs have the same period.

D.

None of the graphs have the same period.

Solution

Correct answer: A

The graphs of $f(\theta)$ and $g(\theta)$ have the same period.

Question 21

Which of the following represents the general solution to the equation $\tan \theta = -1$?

A.

$$\theta = \frac{\pi}{4} + 2k\pi, k \in \mathbb{I}$$

B.

$$\theta = \frac{\pi}{4} + k\pi, k \in \mathbb{I}$$

C.

$$\theta = \frac{3\pi}{4} + 2k\pi, k \in \mathbb{I}$$

D.

$$\theta = \frac{3\pi}{4} + k\pi, k \in \mathbb{I}$$

Solution

Correct answer: D

$$\theta = \frac{3\pi}{4} + k\pi, k \in \mathbb{I}$$

Question 22

1 mark

If $(3, -2)$ is a point on the graph of $y = f(x)$, what point must be on the graph of $y = 2f(x + 1)$?

A.

$(4, -1)$

B.

$(4, -4)$

C.

$(2, 1)$

D.

$(2, -4)$

Solution

Correct answer: D

$(2, -4)$

Question 23

Which equation is represented by the graph sketched below?

A.

$$y = \left(\frac{1}{2}\right)^{-x}$$

B.

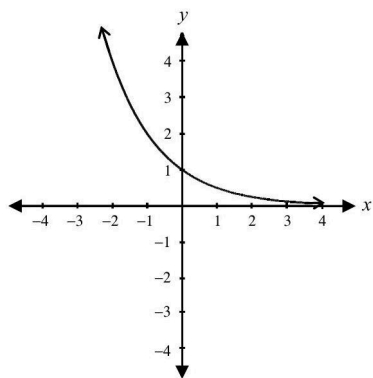
$$y = \left(\frac{1}{2}\right)^x$$

C.

$$y = 2^x$$

D.

$$y = -2^x$$



Solution

Correct answer: B

$$y = \left(\frac{1}{2}\right)^x$$

Question 24

1 mark

What is the degree of the polynomial represented below?

A.

2

B.

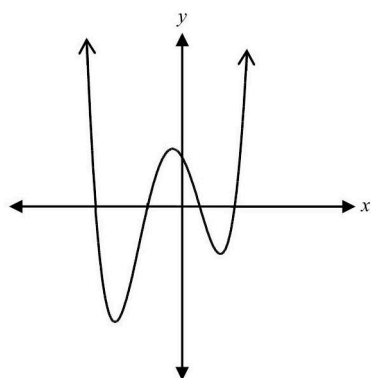
3

C.

4

D.

5



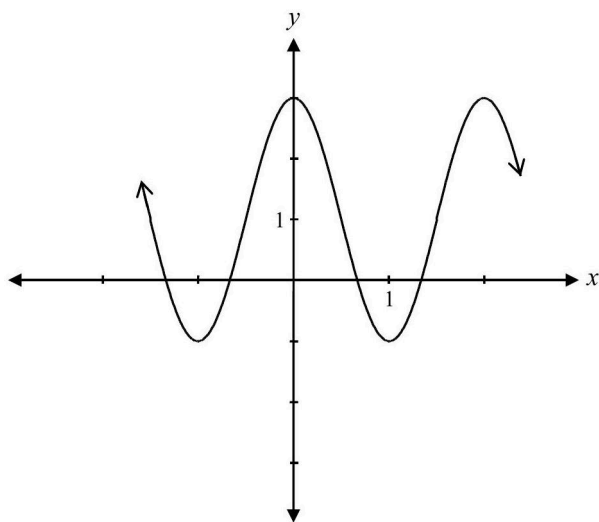
Solution

Correct answer: C

4

Question 25

Given the graph of $y = 2 \cos \pi x + 1$ below, determine another equation that will produce the same graph.



$y =$

Solution

Some sample equations are:

$$y = 2 \cos \pi(x - 2) + 1$$

$$y = -2 \cos \pi(x - 1) + 1$$

$$y = -2 \cos \pi(x + 1) + 1$$

$$y = 2 \sin \pi\left(x + \frac{1}{2}\right) + 1$$

$$y = 2 \sin \pi\left(x - \frac{3}{2}\right) + 1$$

Other answers are possible.

Question 26

Given $f(x) = 3$ and $g(x) = x + 2$, determine the domain and range of $h(x) = \frac{f(x)}{g(x)}$.

Domain:

Range:

Solution

Domain: $\{x \mid x \in \mathbb{R}, x \neq -2\}$

Range: $\{y \mid y \in \mathbb{R}, y \neq 0\}$

1 mark for domain

1 mark for range

2 marks

Explain how to find the exact value of $\sec\left(\frac{19\pi}{6}\right)$.

Find the exact value of $\cos\left(\frac{19\pi}{6}\right)$.

Then take the reciprocal of the value of $\cos\left(\frac{19\pi}{6}\right)$.

1 mark for $\cos\left(\frac{19\pi}{6}\right)$

1 mark for reciprocal

2 marks

Question 27

Explain how to find the exact value of $\sec\left(\frac{19\pi}{6}\right)$.

Solution

Find the exact value of $\cos 6\pi \times \frac{5}{6}$. Then take the reciprocal of the value of $\cos 6\pi \times \frac{5}{6}$. 1 mark for domain 1 mark for range 2 marks 1 mark for $\cos 6\pi \times \frac{5}{6}$ 1 mark for reciprocal 2 marks

Question 28

Given $f(x) = 4 - x$, verify that $f^{-1}(x) = f(x)$.

Solution

Method 1

$$y = 4 - x$$

To find $f^{-1}(x)$, switch x and y values.

$$x = 4 - y$$

$$-y = x - 4$$

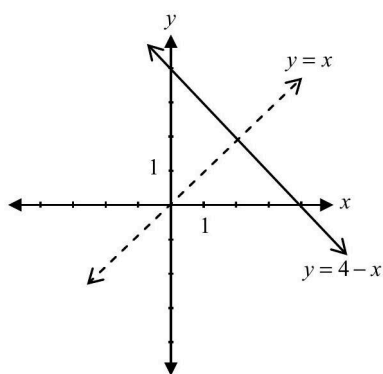
$$y = 4 - x$$

$$f^{-1}(x) = 4 - x$$

1 mark for verifying $f^{-1}(x) = f(x)$

1 mark

Method 2



When $y = 4 - x$ is reflected over the line $y = x$ it produces the same graph.

1 mark

Method 3

Assume $f^{-1}(x) = 4 - x$.

$$\begin{aligned} f(f^{-1}(x)) &= 4 - (4 - x) \\ &= x \end{aligned}$$

$\therefore f(x)$ and $f^{-1}(x)$ are inverses of one another.

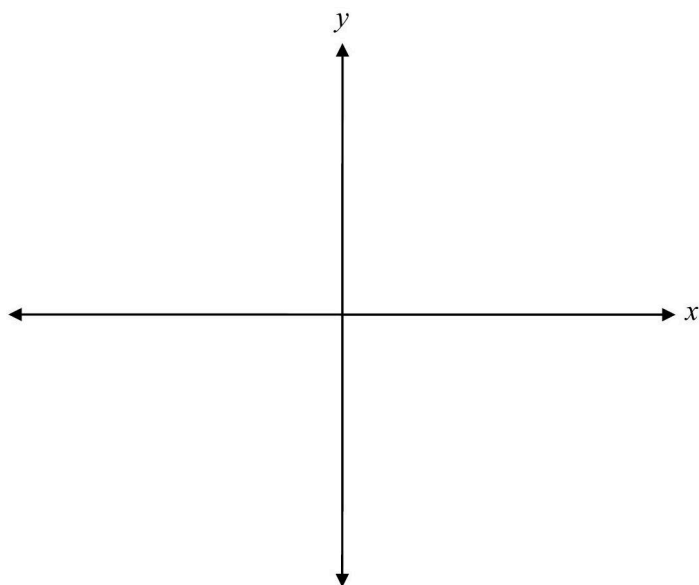
1 mark

Question 29

Sketch the graph of:

$$f(x) = (2 - x)(x + 3)(x + 1)^2$$

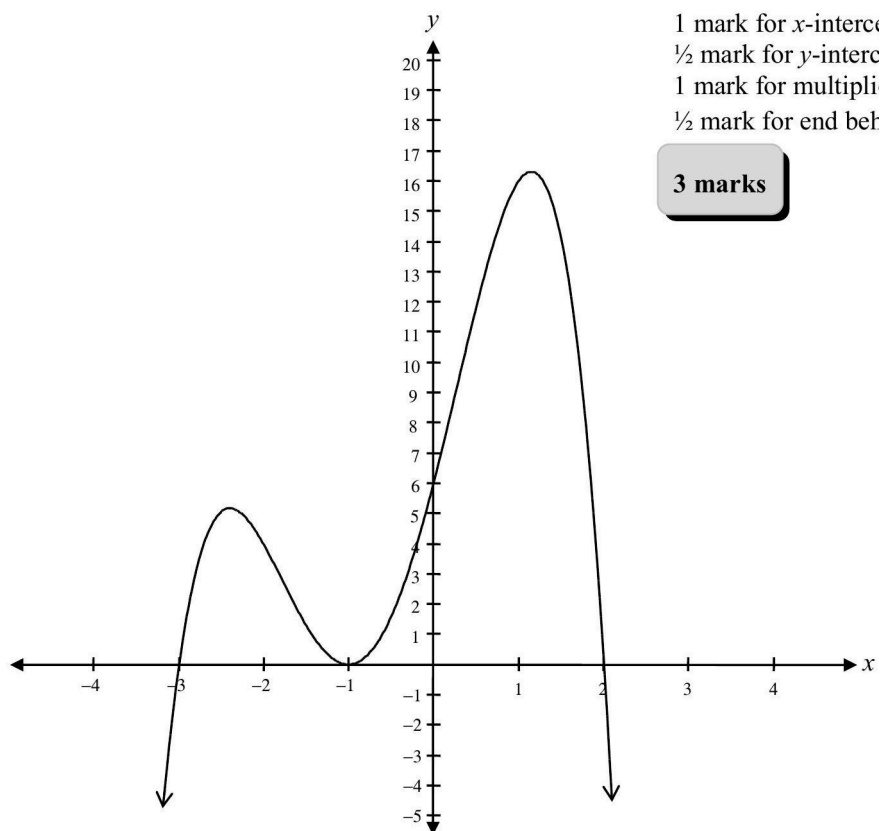
Label the x -intercepts and y -intercept.



Solution

x -intercepts: -3 , -1 , and 2

y -intercept: 6



Question 30

Which expression has a larger value?

$$\log_2 36 \quad \text{or} \quad \log_3 80$$

Justify your answer.

Solution

Method 1

$$\log_2 36 \quad 2^? = 36 \quad \begin{array}{l} 2^5 = 32 \\ 2^6 = 64 \end{array} \quad \approx 5.1$$

$$\log_3 80 \quad 3^? = 80 \quad \begin{array}{l} 3^3 = 27 \\ 3^4 = 81 \end{array} \quad \approx 3.9$$

$\therefore \log_2 36$ is the larger value

1 mark for justification

1 mark

Method 2

$\log_2 32 = 5 \therefore \log_2 36$ is a little more than 5

$\log_3 81 = 4 \therefore \log_3 80$ is a little less than 4

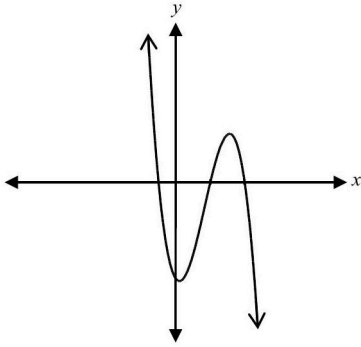
$\therefore \log_2 36$ is the larger value

1 mark for justification

1 mark

Question 31

The graph below represents the equation $y = ax^3 + 6x^2 + 5x - 10$.



What must be true about the value of a ? Explain your reasoning.

Solution

a is any negative number.

Explanation with reference to end behaviour.

or

a cannot be zero.

The graph is of a cubic function,

not a quadratic function.

$\frac{1}{2}$ mark

$\frac{1}{2}$ mark for explanation

1 mark

$\frac{1}{2}$ mark

$\frac{1}{2}$ mark for explanation

1 mark

Question 32

The terminal arm of an angle θ , in standard position, intersects the unit circle in Quadrant IV at a point P $\left(\frac{\sqrt{5}}{4}, y\right)$. Determine the value of $\sin \theta$.

Solution

Method 1

The point P(θ) on the unit circle has coordinates ($\cos \theta, \sin \theta$).

$$\cos^2 \theta + \sin^2 \theta = 1 \quad 1/2 \text{ mark for showing } y = \sin \theta$$

$$\left(\frac{\sqrt{5}}{4}\right)^2 + \sin^2 \theta = 1 \quad 1/2 \text{ mark for substitution}$$

$$\sin^2 \theta = 1 - \frac{5}{16}$$

$$\sqrt{\sin^2 \theta} = \sqrt{\frac{11}{16}}$$

$$\sin \theta = \pm \frac{\sqrt{11}}{4} \quad 1/2 \text{ mark for solving for } \sin \theta$$

$$\sin \theta = -\frac{\sqrt{11}}{4} \quad 1/2 \text{ mark for a negative } \sin \theta \text{ value in Quadrant IV}$$

Method 2

$$(\sqrt{5})^2 + y^2 = 4^2 \quad 1/2 \text{ mark for substitution}$$

$$5 + y^2 = 16$$

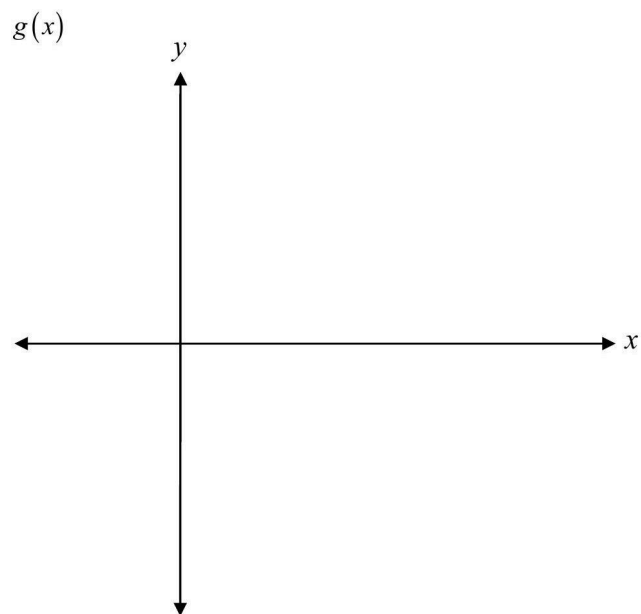
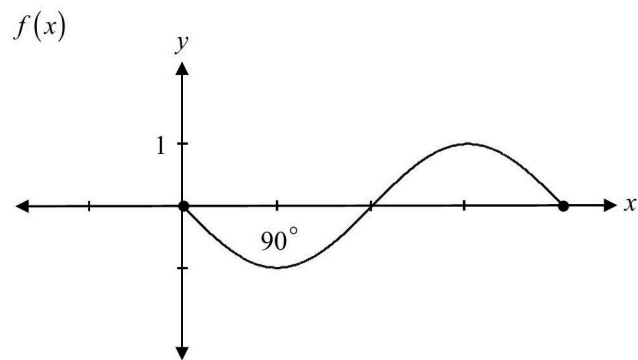
$$y^2 = 11 \quad 1/2 \text{ mark for solving for } y$$

$$y = \pm\sqrt{11} \quad 1/2 \text{ mark for using the value of } y \text{ to find the value of } \sin \theta$$

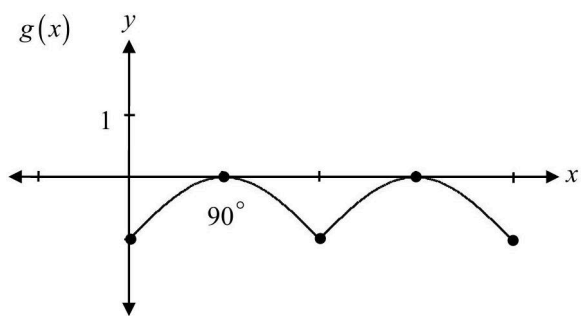
$$\sin \theta = -\frac{\sqrt{11}}{4} \quad 1/2 \text{ mark for a negative } \sin \theta \text{ value in Quadrant IV}$$

Question 33

Given the sinusoidal function $f(x)$ below, sketch the graph of $g(x) = |f(x)| - 1$.



Solution



1 mark for absolute value

1 mark for vertical shift

2 marks

Question 34

The graph of a rational function, $f(x)$, has a point of discontinuity when $x = 2$ and an asymptote when $x = 4$. Write a possible equation for $f(x)$.

Solution

A possible equation is:

$$f(x) = \frac{x - 2}{(x - 2)(x - 4)}$$

1 mark for $\frac{x-2}{x-2}$ (point of discontinuity when $x = 2$)

1 mark for $x - 4$ in denominator (asymptote when $x = 4$)

2 marks

Question 35

Given that $(x - 1)$ is one of the factors, express $x^3 - 57x + 56$ as a product of factors.

Solution

1

1	0	-57	56
↓	1	1	-56
1	1	-56	0

$(x - 1)(x^2 + x - 56)$

or

$(x - 1)(x + 8)(x - 7)$

$\frac{1}{2}$ mark for $x = 1$

1 mark for synthetic division (or for any other equivalent strategy)

$\frac{1}{2}$ mark for consistent factors

2 marks

Question 36

Give an example using values for A and B , in degrees or radians, to verify that

$\cos(A + B) = \cos A + \cos B$ is not an identity.

Left-Hand Side	Right-Hand Side

Solution

Method 1

Let $A = 45^\circ$ and $B = 90^\circ$.

LHS	RHS
$\cos(45^\circ + 90^\circ)$	$\cos 45^\circ + \cos 90^\circ$
$\cos(135^\circ)$	$\cos 45^\circ + \cos 90^\circ$
$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2} + 0$
$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

1 mark for simplification of $\cos(A + B)$

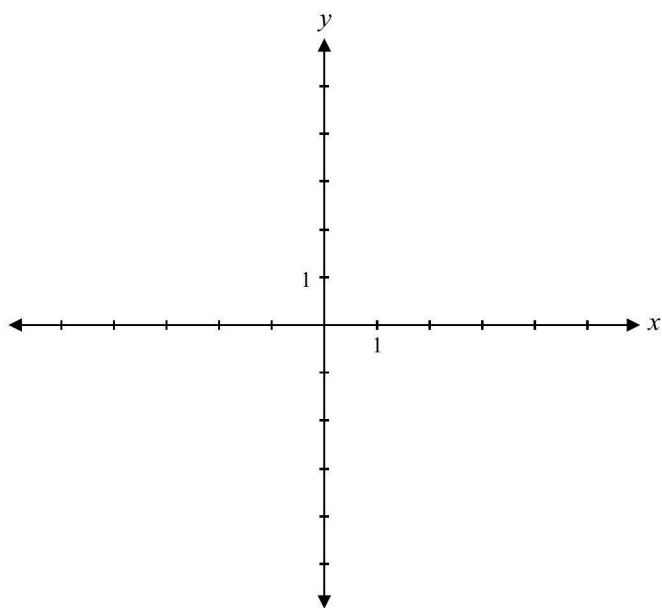
1 mark for simplification of $\cos A + \cos B$

2 marks

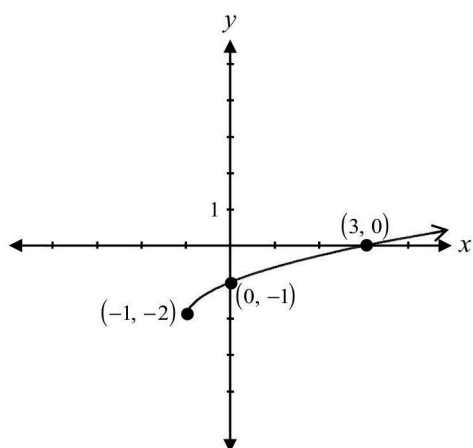
$\text{LHS} \neq \text{RHS} \therefore \cos(A + B) = \cos A + \cos B$ is not an identity.

Question 37

Sketch the graph of $y = \sqrt{x+1} - 2$ and verify that the value of the x -intercept is the same as the solution to the equation $\sqrt{x+1} - 2 = 0$.



Solution



1 mark for general shape

$\frac{1}{2}$ mark for horizontal shift

$\frac{1}{2}$ mark for vertical shift

$$\begin{aligned}\sqrt{x+1} &= 2 \\ (\sqrt{x+1})^2 &= (2)^2 \\ x+1 &= 4 \\ x &= 3\end{aligned}$$

$$\begin{aligned}\sqrt{x+1} - 2 &= 0 \\ \sqrt{3+1} - 2 &= 0 \\ \sqrt{4} - 2 &= 0 \\ 0 &= 0\end{aligned}$$

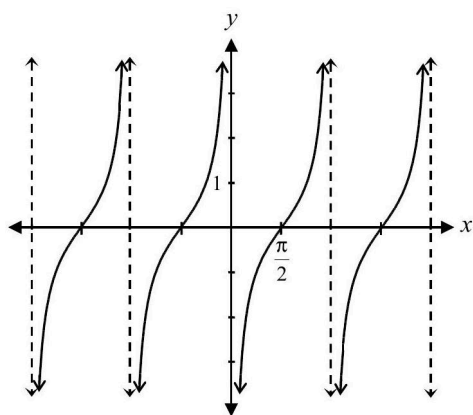
1 mark for verification

3 marks

Question 38

Mohamed is asked to sketch the graph of $y = \tan x$.

His graph is shown below.



Explain why his graph is incorrect.

Solution

The graph of $y = \tan x$ should have zeros at $k\pi, k \in \mathbb{I}$.

1 mark for explanation

or

1 mark

The graph of $y = \tan x$ should have asymptotes at $(2k + 1)\frac{\pi}{2}$ or $\frac{\pi}{2} + k\pi, k \in \mathbb{I}$.

or

Mohamed sketched the incorrect graph. He sketched the graph of $y = \tan(x - \frac{\pi}{2})$.

Question 39

On the interval $0 \leq \theta < 2\pi$, identify the non-permissible values of θ for the trigonometric identity:

$$\tan \theta = \frac{1}{\cot \theta}$$

Solution

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{\frac{\cos \theta}{\sin \theta}}$$

$\therefore$ the above identity is non-permissible

when $\cos \theta = 0$ or $\sin \theta = 0$.

1 mark for identifying non-permissible values

($\frac{1}{2}$ mark for $\cos \theta = 0$, $\frac{1}{2}$ mark for $\sin \theta = 0$)

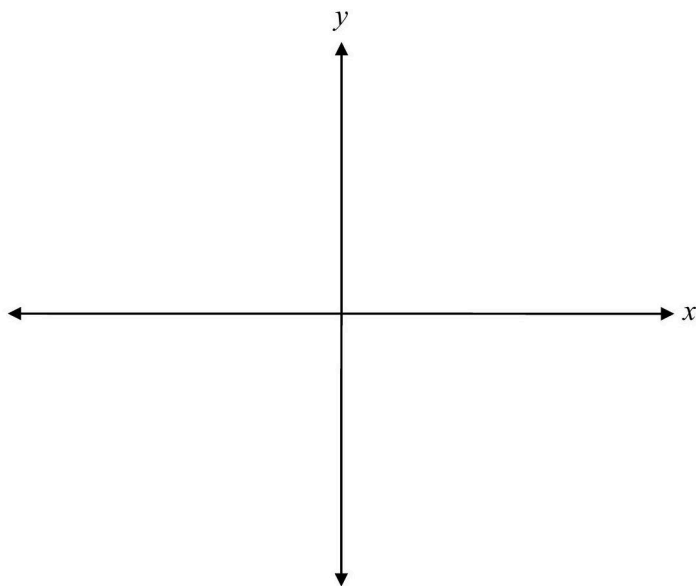
$$\begin{array}{lcl} \cos \theta & \neq & 0 \\ \theta & \neq & \frac{\pi}{2}, \frac{3\pi}{2} \\ \theta & \neq & 0, \pi, \frac{3\pi}{2} \end{array} \quad \begin{array}{l} \sin \theta \neq 0 \\ \theta \neq 0, \pi \end{array}$$

1 mark for solving for θ ($\frac{1}{2}$ mark for each solution set)

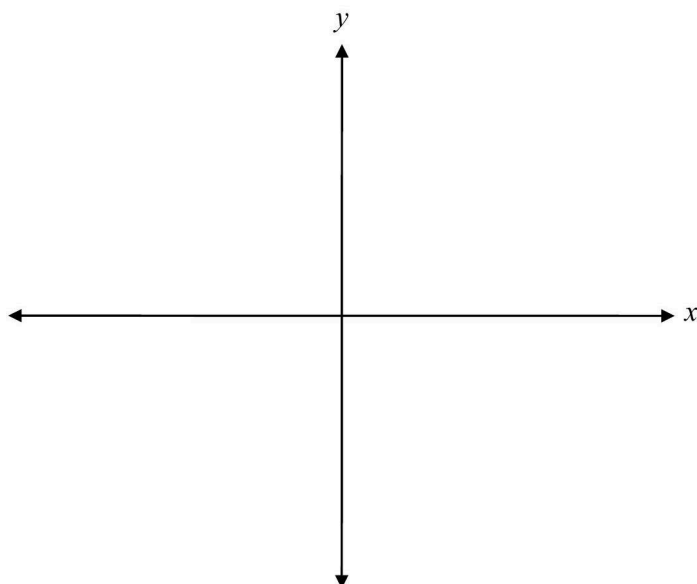
2 marks

Question 40

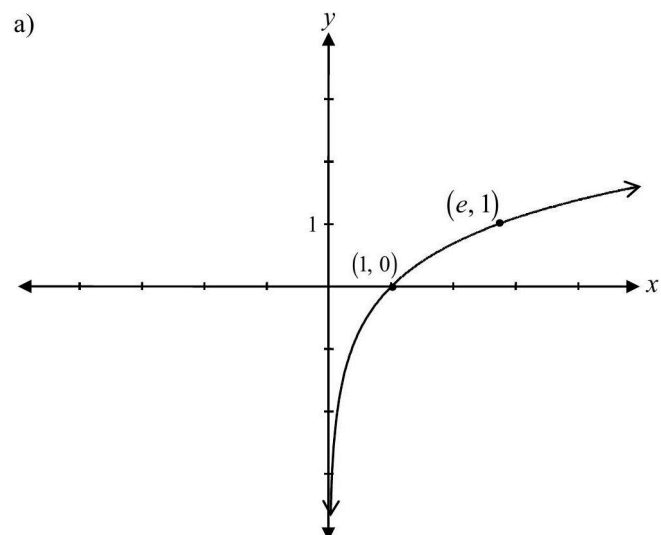
a) Sketch the graph of $y = \ln(x)$.



b) Sketch the graph of $y = -\ln(x - 2)$.



Solution



$\frac{1}{2}$ mark for increasing logarithmic

function

$\frac{1}{2}$ mark for x -intercept at $(1, 0)$

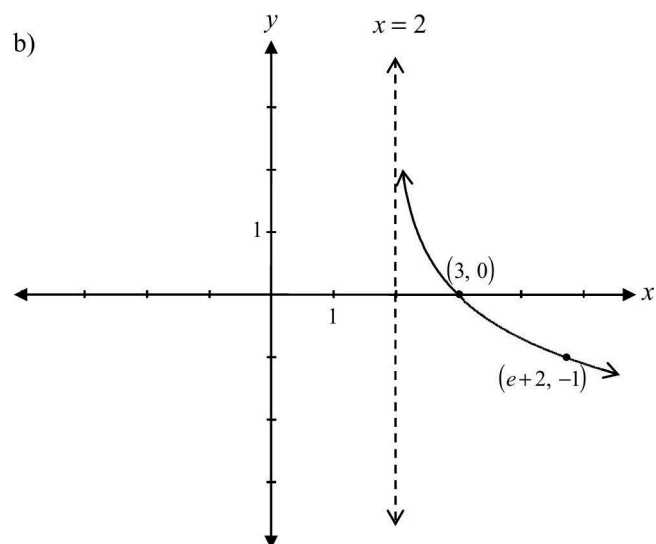
$\frac{1}{2}$ mark for consistent point on

logarithmic function

$\frac{1}{2}$ mark for vertical asymptotic

behaviour

2 marks



1 mark for reflection in x -axis

1 mark for horizontal shift

2 marks

Question 41

Given $f(x) = \sqrt{x-2}$ and $g(x) = 3x$, write the equation for $h(x) = f(g(x))$.

What are the restrictions on the domain of $h(x)$?

Explain your reasoning.

Solution

$$h(x) = \sqrt{3x-2}$$

$$3x - 2 \geq 0$$

$$3x \geq 2$$

$$x \geq \frac{2}{3}$$

Since we cannot find a square root of
a negative number, there is a restriction
on the domain, $x \geq \frac{2}{3}$.

1 mark for $h(x) = f(g(x))$

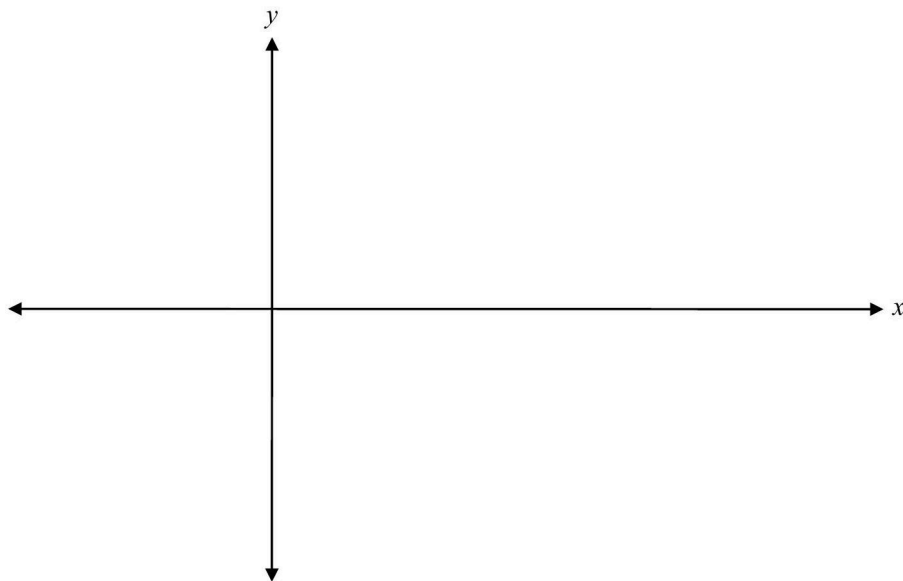
$\frac{1}{2}$ mark for identifying restriction

$\frac{1}{2}$ mark for explanation

2 marks

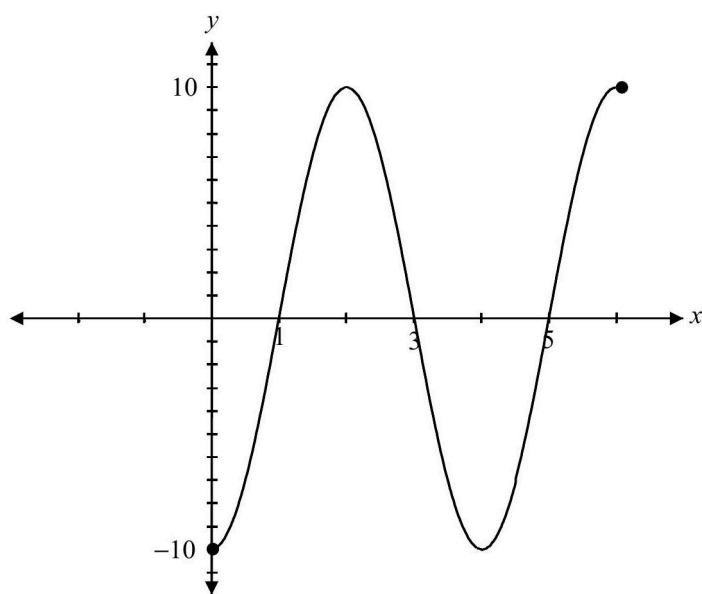
Question 42

Sketch the graph of $y = 10 \cos\left[\frac{\pi}{2}(x - 2)\right]$ over the interval $[0, 6]$.



Solution

$$\text{period} = \frac{2\pi}{\frac{\pi}{2}} = 4$$



1 mark for amplitude

1 mark for period

1 mark for horizontal shift

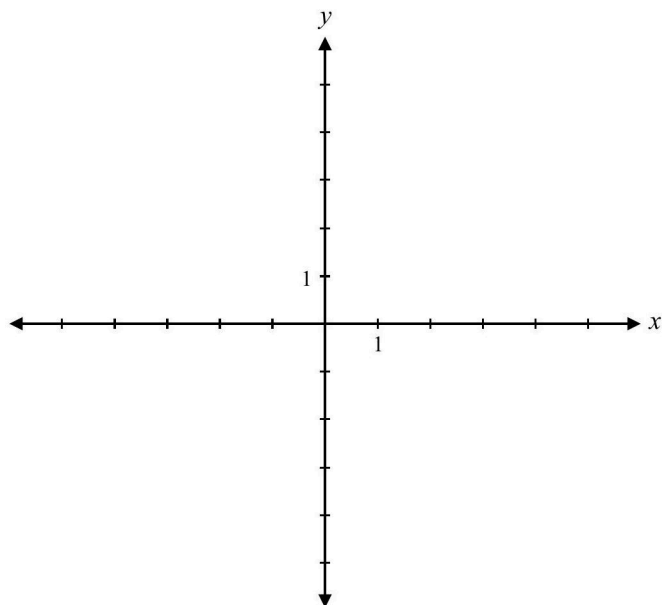
3 marks

Note(s):

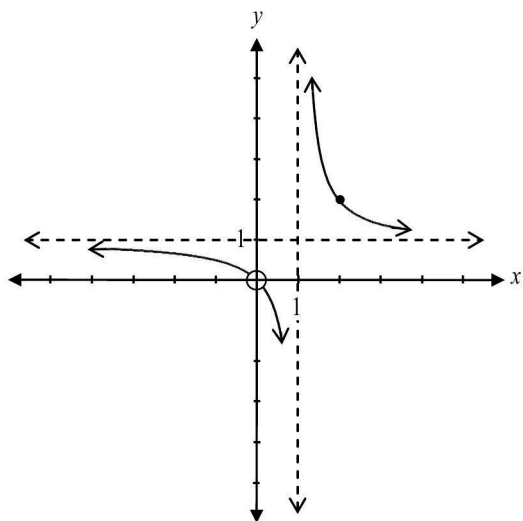
- deduct $\frac{1}{2}$ mark if the interval $[0, 6]$ is not completely sketched

Question 43

Sketch the graph of the function $f(x) = \frac{x^2}{x^2 - x}$.



Solution



$$f(x) = \frac{x^2}{x(x-1)}$$

$$= \frac{x}{x-1} \text{ with a point of discontinuity where } x = 0$$

point of discontinuity: $f(0) = \frac{0}{0-1} = 0$

$\therefore$ there is a point of discontinuity at $(0, 0)$.

divide:

$$\begin{array}{r} x-1 \overline{) x+0} \\ \underline{x-1} \\ 1 \end{array}$$

or

$$\begin{aligned} f(x) &= \frac{x}{x-1} \\ &= \frac{x-1+1}{x-1} \\ &= 1 + \frac{1}{x-1} \end{aligned}$$

1 mark for vertical asymptote at $x = 1$

1 mark for horizontal asymptote at $y = 1$

1 mark for point of discontinuity at $(0, 0)$ or a point of discontinuity consistent with graph

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

4 marks

Question 44

Is $(x - 3)$ a factor of $x^4 - x^3 - 3x^2 + x - 1$?

Justify your answer.

Solution

Method 1

$$\begin{aligned}x &= 3 \\ \therefore (3)^4 - (3)^3 - 3(3)^2 + (3) - 1 &= 81 - 27 - 27 + 3 - 1 \\ &= 29\end{aligned}$$

The remainder does not equal zero,

therefore $(x - 3)$ is not a factor.

$\frac{1}{2}$ mark for $x = 3$

1 mark for remainder theorem

$\frac{1}{2}$ mark for explanation

2 marks

Method 2

3]

1	-1	-3	1	-1
↓	3	6	9	30
1	2	3	10	29

The remainder does not equal zero,

therefore $(x - 3)$ is not a factor.

$\frac{1}{2}$ mark for $x = 3$

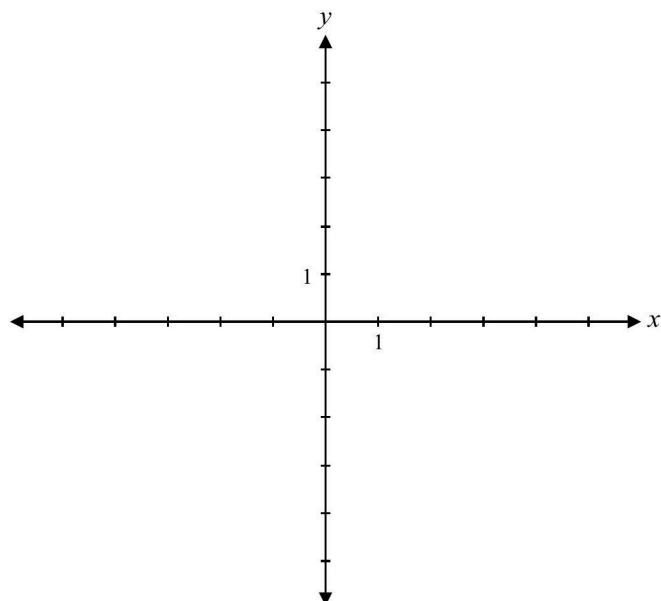
1 mark for synthetic division

$\frac{1}{2}$ mark for explanation

2 marks

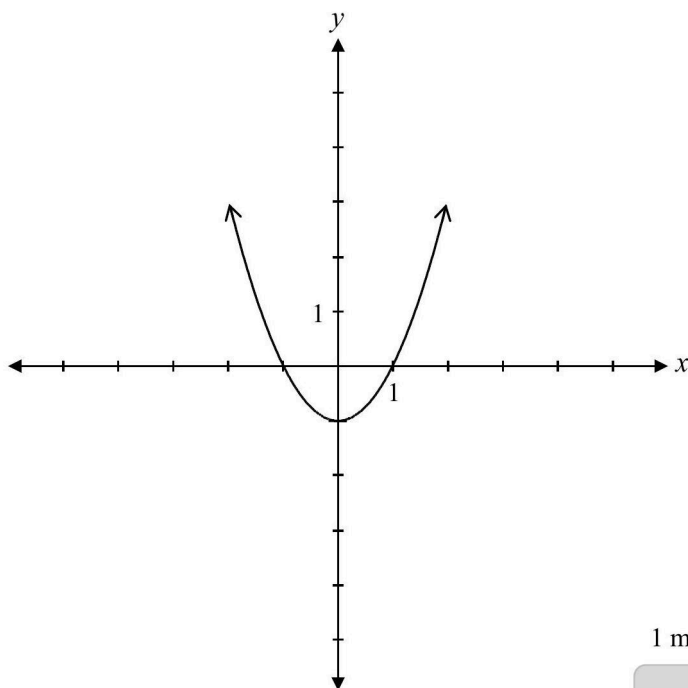
Question 45

Given $f(x) = x - 1$ and $g(x) = x^2$, write the equation of $y = f(g(x))$ and sketch the graph.



Solution

$$f(g(x)) = x^2 - 1 \quad 1 \text{ mark for composition}$$



1 mark for consistent graph

2 marks